

LINCOLN COUNTY AP, NC, USA

WMO: 722128

Lat: 35.483N

Lon: 81.161W

Elev: 267

StdP: 98.16

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 53899

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.3	-22.5	0.5	3.7	-17.9	0.8	2.1	8.3	9.8	7.4	10.0	0.4	300	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.5	22.6	33.1	22.4	32.3	22.2	25.7	30.1	24.8	29.6	24.0	28.6	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.8	27.1	23.5	18.9	26.4	22.7	18.0	25.6	80.7	30.9	76.6	29.4	73.7	28.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.5	4.7	-12.6	36.3	2.9	1.5	-14.7	37.3	-16.4	38.2	-18.0	39.0	-20.2	40.1		
(5)			-13.3	25.8	2.5	1.4	-15.1	26.9	-16.5	27.7	-18.0	28.5	-19.8	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	4.7	6.9	10.4	15.3	19.8	24.4	25.9	25.4	22.4	15.7	9.5	6.6		
	DBStd	8.56	5.49	5.06	5.23	4.31	3.45	2.30	2.03	2.33	2.99	4.36	4.55	4.75		
	HDD10.0	559	179	112	62	8	0	0	0	0	0	6	64	128		
	HDD18.3	1870	422	321	250	109	27	1	0	0	6	104	267	363		
	CDD10.0	2611	16	25	74	166	304	431	492	477	371	183	48	23		
	CDD18.3	879	0	1	4	17	73	181	234	218	127	23	1	1		
	CDH23.3	8132	0	6	62	233	715	1679	2209	1949	1055	208	12	2		
	CDH26.7	3356	0	0	8	44	222	722	1034	874	413	38	1	0		
(14) Wind	WSAvg	1.3	1.5	1.7	1.9	2.0	1.3	1.1	1.0	0.9	1.1	1.2	1.2	1.2		
(15) Precipitation	PrecAvg	1106	98	81	109	86	85	113	94	110	92	86	74	84		
	PrecMax	1597	159	148	223	231	259	221	218	231	195	324	152	148		
	PrecMin	776	27	33	34	15	36	35	48	14	10	0	19	36		
	PrecStd	203	38	32	47	50	53	55	39	65	44	75	39	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	24.1	27.8	29.8	32.5	35.8	36.1	37.0	33.9	29.6	25.1	22.5		
		MCWB	14.5	16.1	15.3	16.6	20.3	22.7	23.0	21.2	19.7	19.7	16.0	15.7		
	2%	DB	18.7	21.2	25.0	27.8	31.0	33.8	34.3	34.1	32.5	27.6	22.5	19.6		
		MCWB	13.6	13.8	14.8	15.9	20.1	22.0	23.4	22.6	20.4	18.6	15.0	15.8		
	5%	DB	17.0	18.5	22.7	26.3	29.1	32.4	33.0	32.7	31.2	26.0	20.6	17.4		
		MCWB	12.8	12.8	13.7	15.7	19.4	22.0	23.2	22.8	20.8	17.8	14.8	13.9		
	10%	DB	14.0	16.3	20.1	24.0	27.5	31.0	32.1	31.3	29.1	23.7	18.3	15.2		
		MCWB	10.1	10.4	12.6	14.9	18.8	21.7	23.0	22.9	20.7	17.8	13.6	11.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.8	18.4	20.4	23.3	25.4	26.7	27.5	25.5	23.0	19.8	19.1		
		MCDB	18.1	21.2	22.0	24.0	28.2	30.0	30.8	30.2	28.3	24.6	21.6	20.9		
	2%	WB	15.5	16.5	16.9	19.2	22.2	24.4	25.6	26.1	23.6	22.1	17.9	17.3		
		MCDB	18.0	18.8	21.3	23.5	26.9	29.3	30.8	29.7	27.2	24.4	20.0	18.5		
	5%	WB	13.4	14.4	15.6	18.0	21.3	23.7	24.8	24.8	22.8	20.5	16.4	15.1		
		MCDB	15.9	18.2	20.7	22.0	25.8	28.7	30.1	29.3	26.6	23.2	18.8	16.7		
	10%	WB	11.2	11.7	14.2	17.1	20.5	22.9	24.0	23.9	22.1	19.1	14.8	12.4		
		MCDB	13.6	14.5	18.7	21.3	24.9	28.0	29.1	28.1	25.7	22.3	17.5	14.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.4	13.3	13.7	12.9	12.1	11.7	11.3	11.6	13.2	13.9	11.7		
		MCDBR	14.5	15.3	15.9	16.2	15.1	14.6	13.1	13.2	14.8	15.1	16.2	13.8		
	5% WB	MCWBR	9.5	9.3	7.6	6.8	5.7	4.9	4.1	4.3	4.8	7.0	9.3	9.2		
		MCWBR	11.6	13.3	13.2	12.1	12.1	12.4	11.7	11.0	11.2	10.0	11.2	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.320	0.357	0.400	0.453	0.481	0.529	0.516	0.421	0.377	0.328	0.312		
	taud		2.495	2.464	2.398	2.278	2.163	2.122	2.015	2.032	2.285	2.388	2.485	2.523		
	Ebn at Noon		893	924	913	885	839	812	770	771	835	846	863	869		
	Edh at Noon		83	96	112	132	151	157	173	167	124	102	83	76		
(44) All-Sky Solar Radiation	RadAvg		2.61	3.27	4.32	5.38	5.99	6.38	5.95	5.47	4.63	3.81	2.89	2.24		
	RadStd		0.16	0.38	0.30	0.42	0.49	0.46	0.41	0.26	0.42	0.47	0.24	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+109

Nomenclature: See separate page